

Project Instructions

Objectives

The project will consist of performing a **complete Bayesian statistical analysis** on a dataset. The topic of the project is **open**, and students are expected to choose the objective or objectives they wish to address.

On **ATENEA**, you can find a selection of good projects completed by students in previous years, which may help you understand what is expected for this assignment.

Planning

The project should preferably be completed **in groups of two**, although it may also be done **individually or in groups of three**, provided that this contributes to improving the quality of the work.

Before **Friday, April 10**, one member of each group must post on the **course forum** the names of the group members as well as the **chosen topic**.

During the **last weeks of classes**, each group will give a **presentation of their project in class**. The main goal of this presentation is for you to explain the **approach of your project and any questions you may have**. The presentation is recommended to include at least:

1. Statement of the objectives
2. Description of the data and a brief descriptive analysis
3. Implementation of an initial Bayesian model

This presentation will account for **20% of the project grade**.

Based on the feedback received during the presentation—from both the instructors and your classmates—as well as your own ideas, you should continue developing the project and finally submit a **final report**, which will account for **80% of the project grade**.

Regarding the length of the final report, there are **no strict requirements**, but it is reasonable to expect **around 15 pages**, and in no case can it be more than 25 pages. The report must be submitted in **PDF format**.

In addition, you must also submit the **files containing the scripts used to perform all calculations** (R, Stan, etc.) as well as the **data files**.

Both in the presentations and in the final report you can choose the language: **Catalan, Spanish or English**.

The **deadline for submitting the final report is June 9**.

Suggested Structure of the Final Report

The following structure is **only a guideline**. You are **not required to strictly follow it**.

1. Abstract

Brief summary of the problem, analysis, and conclusions (**maximum 10 lines**).

2. Introduction

Description of the problem and justification for its selection.

Objectives, scope of the project, and other relevant information.

Response variable and explanatory variables.

Data presentation: a table with a small sample of the data used.

3. Analysis

Exploratory data analysis.

Bayesian analysis (presentation of the models, model validation, interpretation of the selected final model, prediction, etc.).

4. Conclusion

Main conclusions written in language understandable for **non-statisticians**.

Difficulties encountered.

Possible extensions of the analysis.

5. References